

Herald



Herald — Status

Field	Value
Revision	16
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Status	active

Status
summary

r16 captures Wave 6 (pherald inbound runtime + Claude Code headless bridge — code-doc closure). HRD-100 closed atomically with 12 Wave-6 commit SHAs. Tag v0.4.0 is DEFERRED to T13b (operator-supplied live evidence per §107.x). New Cobra subcommand `pherald listen` runs the closed-loop runtime: Telegram `getUpdates` long-poll (`OnText/OnPhoto/OnDocument/OnVoice`) + bot self-filter (`bot.Me.Username` anti-echo guard) + sha256 content-addressed attachment download into `~/herald/inbox/<sha256>.<ext>` + Claude Code dispatch with Opus pinned in `argv` (`claude --model claude-opus-4-7 --resume <UUID> --print <envelope>`) + envelope pre-text (verbatim operator wording "We have received new message from our communication channel ...") preceding the `<<<HERALD-DISPATCH-v1>>>` structured block + `<<<HERALD-REPLY>>>` parser with `action` routing + `tgram.Adapter.SendReply` wiring `reply_to_message_id.T10a --qa-out-dir JSONL` journaling. 12 commits T1=ad87d7f..T12=96c7c6b. 8 new e2e invariants E63-E70 (all currently SKIP-with-documented-reason — convert to PASS at T10b live evidence). Wave 6 mutation gate tests `/test_wave6_mutation_meta.sh` (3 paired hermetic). Pre-existing e2e FAILs unrelated to Wave 6 (Postgres SASL on :24100 surfaced in E7-E12 + E14-E16 + E37-E42) tracked separately — NOT Wave 6 regressions. Workspace at 16 modules (Wave 5 added `qaherald`; Wave 6 added NO new module — inbound is `pherald/internal/inbound/`). Prior r15 captured Wave 4b TOON (tag v0.3.0). Prior r14 captured Wave 4a HTTP/3+Brotli+Alt-Svc+TLS (tag v0.2.0). r13 closed HRD-016 (Wave 3b `pherald Runner` live). r12 closed HRD-011 (Telegram live, `message_id=5`). Every `/v1/*` business route on every serving flavor now honors `Accept: application/toon` → real TOON-encoded body (via `digital.vasic.toon+toon-format/toon-go` upstream wire-up replacing the prior sentinel-error scaffold from the §107 PASS-bluff revert). JSON remains the honest fallback. `POST /v1/events` accepts both TOON and JSON request bodies (TOON transcoded to JSON before the Runner's `EventParser`; Runner stays codec-agnostic).

Field	Value
	HERALD_DEFAULT_RESPONSE_CODEC=json operator opt-out. TOONMiddleware auto-wired into cli.RunServe — every flavor binary (pherald/cherald/sherald/bherald/rherald/iherald/scherald) inherits for free. Real compression evidence: sherald /v1/safety_state TOON=215B vs JSON=226B (~5% on a small payload; larger tabular payloads see 30-60% per upstream measurements). 7 new e2e invariants E56-E62 (E62 SKIP-with-reason — encoder-failure fallback unit-tested in commons/cli/toon_test.go). New tests/test_wave4b_mutation_meta.sh (5/5 PASS — M1 strip TOONMiddleware → E57 FAIL; M2 swap toon.Marshal → json.Marshal (the EXACT 2026-05-17 PASS-bluff signature) → E60 FAIL — gate proves wire-byte-0 anti-bluff catches the recurrence; M3 no-op transcodeRequestBody → pherald unit test FAIL; working-tree quiescence guard (lesson from 72e8lab — script greps own working tree for residual mutation markers BEFORE declaring PASS); post-flight green). Workspace module count unchanged at 15 (no new module). Tag v0.3.0. Prior r14 captured Wave 4a (HTTP/3 + Brotli + Alt-Svc + TLS 1.3; tag v0.2.0). r13 closed HRD-016 (Wave 3b pherald Runner live). r12 closed HRD-011 (Telegram live). r10-r11: Wave 3a substrate.
Issues	HRD-008, HRD-015, HRD-018 (in_progress), HRD-019..HRD-027, HRD-029..HRD-056, HRD-081, HRD-085..HRD-090 see Issues.md — 47 open items spanning §42 constitution bindings (HRD-018..027), §43 command-catalogue (HRD-029..056), HRD-081 (containers podman-compose adapter refinement), HRD-085..090 (queue-repository stub-completion surface). No HRDs migrated in r15 (substrate upgrade only — Wave 4b). Prior cycle: HRD-016 (pherald Runner) + HRD-011 (Telegram live) + HRD-028 + HRD-098 + HRD-099 closed in r10-r13.
Issues summary	
Fixed	HRD-001..HRD-007, HRD-009, HRD-009b, HRD-010, HRD-011, HRD-012, HRD-013, HRD-014, HRD-016, HRD-017, HRD-028, HRD-080, HRD-092, HRD-093, HRD-094, HRD-095, HRD-096, HRD-097, HRD-098, HRD-099
Fixed summary	see Fixed.md
Continuation	QA workstream next: Constitutional addition for docs/qa/ evidence mandate + qaherald Go binary driving full pherald  Telegram round-trip automation with attached materials preserved under docs/qa/<run-id>; final close-out is tag v1.0.0-dev-0.0.1 (or canonical-named) release. Carry-over Wave 3c items (HRD-024 iherald paging, HRD-033 destructive-guard body, HRD-018..027 flavor bindings, the rest of the §43 command catalogue) remain queued. Each milestone followed by multi-mirror push + full anti-bluff verification.

Table of contents

- [Project phase](#)
- [Specification](#)
- [Implementation](#)
- [Operations](#)
- [Risk surface](#)

Project phase

Implementation-r1, Foundation complete + Wave 2 flavor scaffolds landed. Spec V3 r8 is active, the Go module foundation compiles, all three Foundation milestones (M1/M2/M3) plus

the Wave 2 flavor-scaffold workstream have landed, and the §107 end-user-usability covenant is now explicit in Herald's three root docs + gate (added 2026-05-20). The remaining cycle is wiring live channel integrations (HRD-011, HRD-016, HRD-024, HRD-028, HRD-098), validating the §26.5 quickstart end-to-end on operator hardware, and implementing the 28 §43 command bodies still stubbed.

Specification

- **Active spec:** docs/specs/mvp/specification.V3.md (Revision 8, ~4600 lines after r8 Wave 2 capture).
- **Archived specs:** V1 + V2 in docs/specs/mvp/archive/ (frozen).
- **Recent sections:** §37 Tracker-doc change events; §38 Workable-item announcement contract; §39 Message presentation + Herald Canonical Template; §40 Documentation + 15 named test challenges; §41 REST API surface (Gin Gonic); §42 Constitution-binding integration; §43 Constitution-derived flavor commands (27 entries → HRD-029..HRD-056) + §43.2 catalogue-check addendum (r8); §44 Foundation implementation contract + §44.M Wave 2 milestone capture (r8); §6.3 Branding extension with per-flavor fields (r8); §18.0 Wave 2 flavor-binary catalogue (r8).
- **Spec-change rule:** §23 + HERALD_CONSTITUTION §106 + I7 gate — green throughout.
- **End-user-usability covenant:** HERALD_CONSTITUTION §107 + I8 gate (added 2026-05-20) — green; paired §1.1 mutation meta-test in tests/test_i8_usability_meta.sh (5/5 PASS).

Implementation

Module	Status	Notes
commons (L0)	✓ landed	§11.0 types + Clock + UUIDv7 + DefaultBranding; tests pass.
commons_prefix	✓ landed	§8.2 algorithm + 5 test functions; tests pass.
commons_constitution	✓ landed (M1)	14 files, ~2.9k LOC — Evaluator + Registry + 12 emit helpers + BundleHash + ModeLadder + ConstitutionStore + MemoryBus + Runner + CloudEvents v1.0 adapter; all green under go test -race.
commons_messaging (L1)	partial	null:// adapter fully working; Telegram + Claude Code stubs in place (HRD-011/012 for live wiring). 9 SQL migrations embedded; pgx pool + RLS-enforcing WithTenantContext live (commit c8168ec); HRD-010 (commits 82ea82d..13cea69) wired queue (digital.vasic.background → pgxTaskRepository) + Redis (digital.vasic.cache) + the pherald migrate up/status/down/validate CLI. The §107 covenant proved its worth here: the E14 RLS falsifiability test caught (and the implementer fixed) a real production RLS-bypass bug where the bootstrap PG superuser silently bypassed FORCE ROW LEVEL SECURITY. HRD-085..090 opened to track the remaining 16 pgxTaskRepository methods returning ErrUnsupported.
commons_storage (L1)	✓ landed (M2 + HRD-010)	

Module	Status	Notes
commons_infra	✅ landed	QuickstartBoot + WithTenantContext + on-demand container orchestration via <code>containers/</code> .
commons/cli (L0)	✅ landed (Wave 2 / HRD-092)	Shared CLI scaffold: NewRootCmd + VersionCmd + ServeCmd (with Middleware hook) + StubCmd + healthz/readyz/metrics handlers. Consumed by pherald + 6 new flavor binaries. Catalogue-check no-match → vendored Herald-internal per §11.4.74.
pherald (CLI)	✅ partial (M3 + Wave 2 refactor)	Cobra root + version + serve (live HTTP on :24791, /v1/healthz, /v1/readyz, /v1/events, /metrics) now consumes cli.ServeCmd with RequestIDMiddleware hook (commits 1f81d69, 31562cf). send/doctor/migrate/subscriber/deadletter stubbed with HRD pointers.
sherald (CLI)	✅ live (Wave 3a / HRD-098)	System Herald binary serving :24793 via cli.ServeCmd + commons_auth.GinMiddleware (JWT gate). /v1/safety_state now LIVE — backed by sherald/internal/safety (process-local Aggregator + 15s mem-sampler goroutine, lifecycle bound to SIGTERM-cancelled context). 5 §43 stubs (HRD-033/034/040/046/056) still return honest non-zero + HRD pointer.
cherald (CLI)	✅ live (Wave 3a / HRD-028)	Constitution Herald binary serving :24792 via cli.ServeCmd + commons_auth.GinMiddleware (JWT gate). /v1/compliance now LIVE — backed by cherald/internal/compliance (paginated + filter-aware ConstitutionStore reads; tenant scope extracted from JWT claims). 11 §43 stubs still return honest non-zero + HRD pointer.
commons_auth	✅ landed (Wave 3a / HRD-099, renumbered from HRD-093)	JWT verifier — HMAC (HS256) + JWKS (RS256/ES256) hybrid; Redis-cached keys with in-memory fallback; Gin middleware + claims helpers. Catalogue-Check: no-match → vendored Herald-internal (<code>docs/catalogue-checks/HRD-099-commons-auth.md</code>).
bherald (CLI)	✅ scaffolded (Wave 2 / HRD-095)	Build Herald CLI-only binary. 3 §43 stubs (HRD-035/041/045).
rherald (CLI)	✅ scaffolded (Wave 2 / HRD-096)	Release Herald CLI-only binary. 3 §43 stubs (HRD-031/032/045).
iherald (CLI)	✅ scaffolded (Wave 2 / HRD-097)	Incident Herald binary serving :24794 via cli.ServeCmd. /v1/webhooks/page returns honest 501 + HRD-024.
scherald (CLI)	✅ scaffolded (Wave 2 / HRD-097)	Scheduled-audit Herald CLI-only binary. 1 §43 stub (HRD-047).

Operations

- **Repo hygiene:** main only; four-mirror fan-out proven across spec + Foundation M1/M2/M3 + anti-bluff infra + §107 covenant landing.
- **Constitution inheritance gate: 15 PASS / 0 FAIL** (was 12; I8a/b/c added 2026-05-20 for §107 covenant).
- **Anti-bluff audit: 16 PASS / 0 FAIL / 1 SKIP** (was 14; HRD-011 telebot.v3 third-party SKIP-with-reason per §11.4.74; I8 paired meta-test included).
- **CodeGraph validate:** 7 PASS / 0 FAIL / 2 SKIP (HRD-091 documented submodule-traversal gap).
- **E2E bluff-hunt against real services: 41 PASS / 0 FAIL / 5 SKIP** (was 33; Wave 3a added 8 new PASSES: E35/E36 auth gate × 2 flavors + E43/E44 cherald compliance live + E46/E47 sherald safety_state live; updated E28/E29 from "501 stub" → "401 JWT-gated" to reflect Wave 3a route swap). SKIPs are E17/E18/E34 (Telegram live, claude live, full inbound — requires operator credentials) + E45 (cross-binary cherald-reads-pherald-writes — Wave 3b once Runner lands) + E48 (sherlad destructive-op trigger — HRD-033 body not implemented). Real Gin server, live /v1/* endpoints, real Postgres + Redis containers via containers/ submodule, M2 RLS integration tests + HRD-010 storage tests against the live stack, SIGTERM-graceful-shutdowns. Paired §1.1 mutation gates: tests/test_wave2_mutation_meta.sh (4/4 PASS — Wave 2 invariants) + tests/test_wave3_mutation_meta.sh (3/3 PASS — Wave 3a invariants M1+M5+post-flight, M6 SKIP). This is the canonical §107 evidence per Herald §107.
- **Submodules:** 12 vendored — 11 Helix-stack capability modules under submodules/ (auth, background, cache, config, database, eventbus, middleware, observability, recovery, plus Models + Concurrency added during HRD-010) referenced via replace directives in consuming go.mod files, NOT via go.work; plus containers/ (runtime auto-detection + on-demand container boot, consumed directly by Foundation tests + pherald doctor). All 12 carry the §11.4 anti-bluff anchor.
- **Build / test toolchain:** Go 1.25+ (verified on 1.26 per CLAUDE.md), Pandoc 3.9, WeasyPrint 66, Podman 5.8 — verified locally.

Risk surface

- **Submodule-catalogue discovery (operator mandate 2026-05-20).** Every new HRD MUST carry a Catalogue-Check: reuse|extend|no-match <org/repo>@<sha> line per Universal §11.4.74. New HRDs (HRD-029..056 §43 catalogue) currently default to TBD pending operator review.
- **Versioning discipline (operator mandate 2026-05-20).** Minor changes bump secondary version, major bump primary. Currently at V3 r7→r8 within the V3 series; a full V4 rewrite is reserved for a primary-version jump.
- **Live channel integrations untested at runtime.** HRD-011 (Telegram telebot live) requires operator-supplied credentials per docs/CONTINUATION.md. HRD-012 (Claude Code claude --resume live) is closed with live PASS evidence captured. The §107 covenant blocks claims of "done" on these until end-user evidence is captured.
- **HRD-008 operator-side quickstart not yet validated end-to-end on a fresh laptop.** The compose stack works in dev; operator hardware validation is the next concrete §107 evidence point.
- **No CI yet.** No GitHub Actions / GitLab CI / etc. Operator runs scripts/e2e_bluff_hunt.sh manually for now — that script IS the local CI surrogate.

Closed risks

- **Production RLS-bypass bug (closed 2026-05-20 during HRD-010 step 3, commit 5c1c022).** Until that commit, `commons_infra.WithTenantContext` set the tenant GUC but `ROW LEVEL SECURITY` was not `FORCED` on the targeted tables — and the bootstrap PG user used in tests was a superuser, which bypasses RLS regardless. The E14 falsifiability test (real Postgres, cross-tenant `SELECT` must return `EXACTLY 0` rows from the other tenant) caught the bluff before any live deployment. Fix: switched the application connection to a non-superuser role + ran `ALTER TABLE ... FORCE ROW LEVEL SECURITY` in migration 000003. The §107 covenant directly prevented this from shipping; the lesson is "EXACT-N row count, never $\geq N$ " for any isolation test.

Recent activity

Wave 3a — substrate + 2 live REST routes (closed 2026-05-21)

8-step workstream commits `dbbea95` . . this shipped:

- `commons_auth/ JWT verifier scaffold` (HRD-099, renumbered from HRD-093; commit `dbbea95`) — new top-level Go module: HMAC (HS256) + JWKS (RS256/ES256) hybrid verifier with Redis-cached keys + in-memory fallback; Gin middleware (`GinMiddleware`) that gates every request behind the verifier and sets `ContextKeyClaims` on success; `claims.go` helpers (`TenantFromClaims`, `SubjectFromClaims`); `verifier_test.go` exercises both modes incl. expired-token + wrong-signature paths. Catalogue-Check: no-match → vendor Herald-internal (`docs/catalogue-checks/HRD-099-commons-auth.md`).
- `events_processed migration` (commit `3f8daed`) — `commons_storage/migrations/000012_events_processed.{up,down}.sql`; RLS-enforced 30-day TTL inbound idempotency archive per V3 §32.2.
- `ConstitutionStore.ListQuery extension` (commit `6276437`) — `Since/Until/Offset` fields added; both Memory + Postgres backends honour them.
- `cherald/internal/compliance/ Handler + tests` (commit `a2ef831`) — paginated + filter-aware `/v1/compliance` handler; tenant scope from JWT claims; defaults `page=1`, `page_size=50` (max 200).
- **cherald wiring + route swap** (HRD-028 close-out, commit `bfca740`) — `cherald/cmd/cherald/main.go` builds the Verifier from env, instantiates the `ConstitutionStore` (Postgres if `HERALD_PG_DSN` else Memory), and wires `commons_auth.GinMiddleware` + `compliance.Handler` via `cli.ServeOpts.Middleware`. The 501-stub for `/v1/compliance` is gone.
- `sherald/internal/safety/ Aggregator + Handler + sampler` (commit `1016b96`) — process-local Aggregator tracks `lastMemPercent` + `lastDestructiveOp` + uptime; `StartMemSampler` goroutine ticks every 15s; Handler returns the snapshot as JSON; tests cover the Aggregator surface.
- **sherald wiring + sampler start** (HRD-098 close-out, commit `bfa82bc`) — `sherald/cmd/sherald/main.go` instantiates the Aggregator + starts the sampler under a SIGTERM-bound context; wires the Verifier + Handler via `cli.ServeOpts.Middleware`. The 501-stub for `/v1/safety_state` is gone.
- **e2e + mutation gate + docs sync (commit 0cc6fad)** — 8 new e2e invariants (E35/E36/E43/E44/E46/E47 PASS; E45/E48 SKIP-with-reason); E28/E29 updated from "501 stub" to "401 JWT-gated"; e2e header set process-wide `HERALD_AUTH_MODE` + `CODEGRAPH_ALLOW_UNSAFE_NODE`; new paired §1.1 mutation gate tests / `test_wave3_mutation_meta.sh` (3/3 PASS); atomic Issues→Fixed for HRD-028,

HRD-098, HRD-099 (commons_auth, renumbered from HRD-093 in this r11 cleanup); Status.md r9 → r10 → r11; Issues.md r8 → r9 → r10; Fixed.md r7 → r8 → r9.

Anti-bluff battery sign-off (this commit): inheritance gate 15/15 PASS; meta-tests (I6, I8, Wave2, Wave3) all green; audit_antibluff.sh 16 PASS / 0 FAIL / 1 SKIP (commons_auth not added to the audit's submodules/ walk — workspace-internal module by design); codegraph_validate.sh 7 PASS / 0 FAIL / 2 SKIP (with CODEGRAPH_ALLOW_UNSAFE_NODE=1 for Node 25 environment); e2e_bluff_hunt.sh 41 PASS / 0 FAIL / 5 SKIP.

Wave 2 — flavor-scaffold sweep (closed 2026-05-21)

15-step workstream commits 7e0a614..eef606b (with parallel logo branding 24b96f2) shipped:

- **Shared scaffold** — new commons/cli/ package: NewRootCmd (steps 2), VersionCmd (step 2), routes.go + healthz/readyz/metrics handlers (step 3), ServeCmd with Middleware-hook + graceful shutdown (step 4), StubCmd honest 501-style stubs (step 1). Vendored Herald-internal per §11.4.74 catalogue-check no-match → see docs/catalogue-checks/HRD-092-commons-cli.md.
- **pherald refactor** — main + stubs migrated to consume commons/cli/ (step 5); pherald_serve consumes cli.ServeCmd with RequestIDMiddleware hook (step 6, commit 31562cf).
- **6 new flavor binaries** — sherald serving :24793 (step 7, HRD-093), cherald serving :24792 (step 8, HRD-094), iherald serving :24794 (step 9, HRD-097), bherald CLI-only (step 10, HRD-095), rherald CLI-only (step 11, HRD-096), scherald CLI-only (step 12, HRD-097). Each ships honest 501-style §43 command stubs with HRD pointers.
- **Anti-bluff additions** — scripts/e2e_bluff_hunt.sh grew from 18 → 33 invariants covering all 6 new binaries (version --json shape, healthz binding, /v1/safety_state + /v1/compliance + /v1/webhooks/page 501 honesty, sherald SIGTERM graceful-shutdown, §43 stub exit-code, pherald regression sentinel). New paired mutation gate tests/test_wave2_mutation_meta.sh (4/4 PASS) proves the new invariants catch their claimed regressions.
- **Spec capture** — V3 r7 → r8 added §6.3 Branding extension (per-flavor fields), §18.0 Wave 2 catalogue, §41 REST route additions, §44.M Wave 2 milestone, §43.2 catalogue-check addendum (step 14, commit eef606b).
- **Atomic Issues→Fixed** — HRD-092..097 closed in r7 of Fixed.md; HRD-098 opened in r8 of Issues.md as Wave 3+ deferred (sherald /v1/safety_state live impl).

Anti-bluff battery sign-off (this commit): inheritance gate 15/15 PASS; meta-tests (I6, I8, Wave2) all green; audit_antibluff.sh 16 PASS / 0 FAIL / 1 SKIP; codegraph_validate.sh 7 PASS / 0 FAIL / 2 SKIP; e2e_bluff_hunt.sh 33 PASS / 0 FAIL / 3 SKIP.